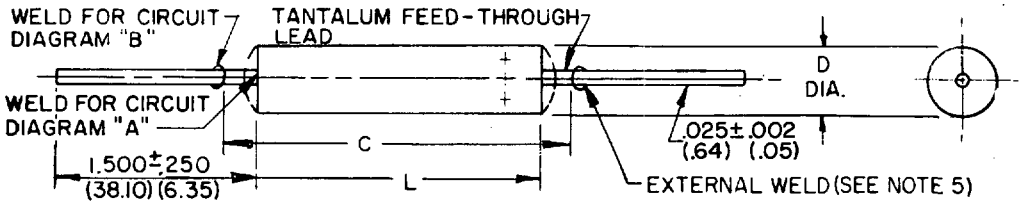
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, ETCHED FOIL) -85°C (VOLTAGE DERATED TO 125°C), STYLES CL20 and CL21

INACTIVE FOR NEW DESIGN; USE MIL-C-39006/1

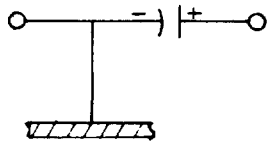
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This Specification is mandatory for use by all Departments and Agencies of the Department of Defense.

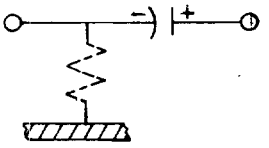


Case Size	Dimensions			
	1/ L ±.062(1.57)	D ±.016(.41)	D Max	C Max
		CL20	CL21	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.469 (37.31)
G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL21, nonshrinkable sleeving shall extend .016 inch minimum, .062 maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.



CASE
CIRCUIT DIAGRAM "A"



CASE
CIRCUIT DIAGRAM "B"
(See Note 4)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL20 and CL21 Capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL2-BE150UPE	15	10	17.2	15	2	4	220	E1
CL2-BE600UPG	15	10	17.2	60	2	10	55	G2
CL2-BE201UPG	15	10	17.2	200	3	40	17	G3
CL2-BE401UPG	15	10	17.2	400	6	80	9	G4
CL2-BE581UPG	15	10	17.2	580	9	100	6	G5
CL2-BG100UPE	25	15	28.8	10	2	4	330	E1
CL2-BG400UPG	25	15	28.8	40	2	13	83	G2
CL2-BG121UPG	25	15	28.8	120	3	40	27	G3
CL2-BG251UPG	25	15	28.8	250	6	80	13	G4
CL2-BG351UPG	25	15	28.8	350	10	100	10	G5
CL2-BH080UPE	30	20	34.5	8	2	6	424	E1
CL2-BH320UPG	30	20	34.5	32	2	14	106	G2
CL2-BH111UPG	30	20	34.5	110	4	50	31	G3
CL2-BH221UPG	30	20	34.5	220	8	90	16	G4
CL2-BH301UPG	30	20	34.5	300	11	120	11	G5
CL2-BJ4R5TPE	50	30	57.5	4.5	2	4	730	E1
CL2-BJ180TPG	50	30	57.5	18	2	13	180	G2
CL2-BJ600TPG	50	30	57.5	60	3	40	55	G3
CL2-BJ101TPG	50	30	57.5	100	4	75	33	G4
CL2-BJ151TPG	50	30	57.5	150	5	100	22	G5
CL2-BL030TPE	75	50	86.2	3	2	4	1000	E1
CL2-BL120TPG	75	50	86.2	12	2	13	270	G2
CL2-BL300TPG	75	50	86.2	30	3	34	110	G3
CL2-BL700TPG	75	50	86.2	70	6	80	47	G4
CL2-BL101TPG	75	50	86.2	100	10	100	33	G5
CL2-BN020SPE	100	65	115.0	2	2	4	1600	E1
CL2-BN080SPG	100	65	115.0	8	2	12	410	G2
CL2-BN250SPG	100	65	115.0	25	2.5	37	130	G3
CL2-BN500SPG	100	65	115.0	50	5	75	66	G4
CL2-BN700SPG	100	65	115.0	70	7	100	47	G5
CL2-BQ010SPE	150	100	172.0	1	2	4	3000	E1
CL2-BQ040SPG	150	100	172.0	4	2	13	750	G2
CL2-BQ130SPG	150	100	172.0	13	3	30	250	G3
CL2-BQ250SPG	150	100	172.0	25	5	80	130	G4
CL2-BQ360SPG	150	100	172.0	36	8	100	94	G5

1/Complete type designation will include an additional symbol to indicate style.

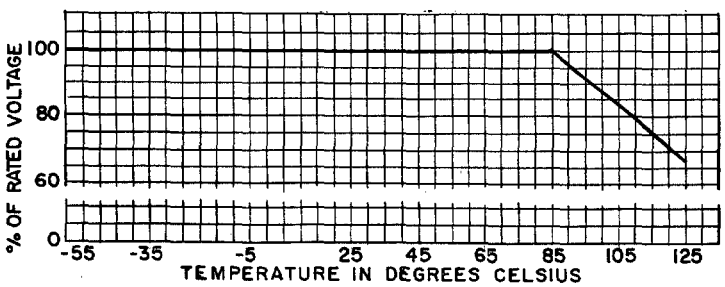


FIGURE 2. Voltage derating with temperature.

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TABLE II SUPERSESSION & CROSS REFERENCE DATA

SUPERSEDED		
Type Designation[1/] MIL-C-3965/17D	Type Designation[1/] MIL-C-3965/17E	Cross Reference For New Design Part Number MIL-C-39006/1C
CL2-BE150UPE	CL2-BE150UPE	M39006/01-2232
CL2-BE600UPE	CL2-BE600UPG	M39006/01-3030
CL2-BE201UPE	CL2-BE201UPG	M39006/01-3031
CL2-BE401UPE	CL2-BE401UPG	M39006/01-3032
CL2-BE581UPE	CL2-BE581UPG	M39006/01-3033
CL2-BG100UPE	CL2-BG100UPE	M39006/01-2237
CL2-BG400UPE	CL2-BG400UPG	M39006/01-3034
CL2-BG121UPE	CL2-BG121UPG	M39006/01-3035
CL2-BG251UPE	CL2-BG251UPG	M39006/01-3036
CL2-BG351UPE	CL2-BG351UPG	M39006/01-3037
CL2-BH080UPE	CL2-BH080UPE	M39006/01-2242
CL2-BH320UPE	CL2-BH320UPG	M39006/01-3038
CL2-BH111UPE	CL2-BH111UPG	M39006/01-3039
CL2-BH221UPE	CL2-BH221UPG	M39006/01-3040
CL2-BH301UPE	CL2-BH301UPG	M39006/01-3041
CL2-BJ4R5TPE	CL2-BJ4R5TPE	M39006/01-2244
CL2-BJ180TPE	CL2-BJ180TPG	M39006/01-3042
CL2-BJ600TPE	CL2-BJ600TPG	M39006/01-3043
CL2-BJ101TPE	CL2-BJ101TPG	M39006/01-3045
CL2-BJ151TPE	CL2-BJ151TPG	M39006/01-3046
CL2-BL030TPE	CL2-BL030TPE	M39006/01-2249
CL2-BL120TPE	CL2-BL120TPE	M39006/01-3047
CL2-BL300TPE	CL2-BL300TPE	M39006/01-3048
CL2-BL700TPE	CL2-BL700TPE	M39006/01-3049
CL2-BL101TPE	CL2-BL101TPG	M39006/01-3050
CL2-BN020SPE	CL2-BN020SPE	M39006/01-2254
CL2-BN080SPE	CL2-BN080SPG	M39006/01-3051
CL2-BN250SPE	CL2-BN250SPG	M39006/01-3052
CL2-BN500SPE	CL2-BN500SPG	M39006/01-3053
CL2-BN700SPE	CL2-BN700SPG	M39006/01-3054
CL2-BQ010SPE	CL2-BQ010SPE	M39006/01-2259
CL2-BQ040SPE	CL2-BQ040SPG	M39006/01-3055
CL2-BQ130SPE	CL2-BQ130SPG	M39006/01-3056
CL2-BQ250SPE	CL2-BQ250SPG	M39006/01-3057
CL2-BQ360SPE	CL2-BQ360SPG	M39006/01-3058

1/ Complete type designation will include an additional symbol to indicate style.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See Fig 1.
Case type - Tubular; uninsulated (CL20), insulated (CL21).
Seal type - Hermetic (symbol G).
- Nonhermetic (symbol E).
Terminals - Axial wire-lead.
Rated temperature - -55°C to +85°C, voltage derated to +125°C (See Fig 2).
Rated voltage - See Table I.
Capacitance (cap) (nom) - See Table I.
Capacitance tolerance - Symbol S(+30, -15%), T(+50, -15%), or U(+75, -15%) as shown in Table I.

DC leakage (DCL)(max)(at +25°C):
See Table I.

Dissipation factor (DF) (max):
20% ($\leq$50v) or 15% ($>$50v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2,000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25°C value of Table I.
ΔCap - Within $\pm$5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: In accordance with MIL-C-3965.

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Terminal Strength:
Pull test - In accordance with MIL-C-3965.
Bend test - Not applicable.
Torque test - Not applicable
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See Table I.

Moisture resistance:
DCL (max) - 125% of 25°C value of Table I.
ΔCap - Within +6% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See Table I (25°C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25°C):
DCL (max) - See Table I.
Cap - Within tolerance of Table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See Table I.
ΔCap (max) - -40% (≤50v) or -35% (>50v) of step 1 value.
Step 3 (+25°C):
DCL (max) - See Table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85°C):
DCL (max) - See Table I.
ΔCap (max) - +20% of step 1 value.
DF (max) - 30% (≤50v) or 20% (>50v).
1/ Step 5 (+125°C):
DCL (max) - See Table I.
ΔCap (max) - +50% (≤30v) or +30% (>30v) of step 1 value.
DF (max) - 30% (≤30v) or 20% (≥30v).

1/ 125°C derated voltage referenced.

Step 6 (+25°C):
DCL (max) - See Table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage:
DCL (max) - 125% of 25°C value of Table I.
ΔCap - Within ±10% of initial measurement.
DF - Shall not exceed initial requirement.

Life at +85°C:
2,000-hr (qualification):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (group B):
DCL (max) - See Table I (25°C):
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
1,760-hr (group C):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF(max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

1/ Life at +125°C:
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF (max) - 20% (≤30v) or 15%(>30v).
DWV (min) (CL21 only) - 2,000 vdc.
IR (min) (CL21 only) - 100 megohms.

Marking:
Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

Voltage groups (85°C rated voltage):
I ---------- 15 to 100 v, incl.
II ---------- 150 v

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/17D. See Table II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review Activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17
DSA - ES

User Activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing Activity:
Air Force - 80

(Project 5910-1048-3)

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